Time period	Technological milestone/development phase	Key innovations introduced	Impact/significance
Early 2010s	Conceptual foundations of cognitive autonomy	Introduction of AI, machine learning (ML), and swarm robotics in UAV research; early automation in flight control systems.	Established the groundwork for self-learning drone behavior and adaptive mission planning.
2015–2020	Transition from fourth to fifth generation	Development of manned–unmanned teaming (MUM-T), digital mission networks, and early combat cloud systems; AI-based mission optimization.	Enabled drones to collaborate with human pilots and other unmanned systems through secure data networks.
2020–2022	Prototype era of cognitive and autonomous UAVs	Programs like Skyborg (United States), Boeing ATS (Australia), and FCAS (Europe)	Marked the first successful demonstration of fully autonomous UAV decision-making and cooperative operations.